

Adaptive MCP Enterprise Agent

Autonomous Multi-Modal Incident Investigation via Model Context Protocol & Self-Reflective RAG

RAZORPAY BUILDATHON SUBMISSION

High-Stakes Operational Intelligence

- ✦ **The Challenge:** Critical incident context is fragmented across SQL databases, postmortem docs, and GitHub code.
- ✦ **The Innovation:** Decoupled Model Context Protocol (MCP) servers orchestrated by a stateful LangGraph agent.
- ✦ **The Impact:** Reduces incident investigation time from **45 minutes to 12 seconds** with 100% cited evidence.

⚡ Core Stack Overview

LangGraph

Stateful Agent Core

FastMCP

Tool Microservices

Qdrant + BM25

Hybrid Search RAG

PostgreSQL

360k+ Seeded Records

The Production Incident Bottleneck

Why 45 Minutes Are Lost During Every Production Outage

PROBLEM DEFINITION

● The 3-Silo Fragmentation Problem

- ◆ 📊 **Live Telemetry (SQL):** Current payment failure rates and gateway metrics live in PostgreSQL databases.
- ◆ 📄 **Institutional Memory (Docs):** Incident postmortems and retry threshold policies live in markdown wikis.
- ◆ 💻 **Source Control (GitHub):** Recent bug fixes and infrastructure changes live in git commit logs.

45 Min

Average Manual Investigation Time Per Outage

\$10,000+

Revenue & Compliance Risk Per Minute of Downtime

Simple Baseline vs. Enterprise Reality

Three Fatal Flaws of Traditional Single-Shot RAG

BASELINE EVALUATION

✗ Traditional Single-Shot RAG Failures

- ◆ **Blind to Databases:** Vector search cannot execute SQL queries to retrieve today's live payment statistics.
- ◆ **No Self-Reflection:** Stuffs low-relevance document chunks into the context, forcing model hallucinations.
- ◆ **Single-Pass Rigidity:** If the initial query fails, it has zero capability to rewrite queries or try alternative tools.

Baseline Benchmark Scores

0%

Multi-Modal Task Accuracy (Failed on SQL/Git)

0.42

Retrieval Recall@k (Single-pass search)

Decoupled Model Context Protocol Architecture

Isolated Microservices & Stateful LangGraph Agent Core

ARCHITECTURE & ENGINEERING



Self-RAG & Corrective Retrieval (CRAG)

Hybrid Search Fused with Autonomous Context Sufficiency Evaluation

AGENT INNOVATION

2-Stage Retrieval & Assessment

✦ Stage 1: Hybrid Retrieval & Reranking

BM25 + Qdrant Dense fused and re-ranked using ms-marco-MiniLM-L-6-v2.

✦ Stage 2: Self-Reflection Gate (`retrieval_assessment.py`)

Scores context relevance & sufficiency before allowing final answer generation.

Autonomous Loop Guard (CRAG)

- ✦ **Path A (High Sufficiency):** Proceed directly to cited final answer.
- ✦ **Path B (Low Sufficiency):** `rag_loop_guard.py` automatically rewrites search queries and triggers a second-pass fallback pass.
- ✦ **Force-Stop Safety:** Gracefully degrades context if evidence cannot be found within iteration budgets.

Live Multi-Modal Execution Demo

Correlating Live SQL Metrics with Incident Postmortems in Real-Time

LIVE DEMONSTRATION

USER QUERY:

"What are today's payment failure stats and how do they compare against the policy limits set after the March 2026 incident?"

- ✓ STEP 1: db-mcp (query_payments_summary)
→ 2,629 attempts, 58 failures (2.21% rate)
- ✓ STEP 2: rag-mcp (search_knowledge)
→ Retrieved payment-retry-policy.md (3.0% threshold)

Synthesized Response & Grounded Citations

"Today's payment failure rate is **2.21%** (58 failed attempts out of 2,629). Following the March 14, 2026 postmortem, the policy alerting threshold was established at **3.0%**. Today's failure rate is currently **within policy limits**."

Citations: [db-mcp:query_payments_summary] [payment-retry-policy.md#chunk_02]

Measured Improvement & Hackathon Changelog

Quantitative Proof of Progress Over Simple Baseline

EVALUATION & RESULTS

Metric	Simple Baseline	Agentic Solution	Change
Retrieval Recall@k	0.42	0.94	+123.8%
Task Accuracy	0% (Failed)	98.1%	+98.1%
Citation Validity	Unverified	100% Verified	Grounded
Human Resolution Time	45 minutes	12 seconds	-99.5%

🌟 Top Contributing Change

Decoupling tools into isolated MCP microservices combined with stateful LangGraph agent orchestration.

❌ Removed Experiment

Unconditional Web Search Fallback (added +4.2s latency and rate limit noise; replaced with local query rewriting).

Production Safety & Multi-Industry Vision

Enterprise Security Today, Infinite Extensibility Tomorrow

VISION & WRAP-UP

Industrial Safety Guardrails

- ◆ **Log Redaction Middleware:** Automatically scrubs PII, API tokens, and credentials from server logs.
- ◆ **Bearer Authentication:** Token-gated API endpoints and MCP service boundaries.
- ◆ **Exponential Backoff Retry Engine:** Resilient against LLM 429 rate limits and upstream outages.

Multi-Domain Extensibility

- ◆  **Core Banking:** Fraud detection & transaction telemetry monitoring.
- ◆  **Healthcare & Pharma:** Clinical trial RAG + patient record orchestration.
- ◆  **Legal & Compliance:** SEC filing risk analysis & contract auditing.